PATHOPHYSIOLOGY — NURSING

Cell Part 2

Cellular Replication & Genetics

Complete exam-detail notes from the PowerPoint lecture recording. Every definition, number, example, mechanism, clinical application, Punnett-square rule, and comparison the instructor flagged is here.

1. Cellular Replication — Overview

Even in healthy people, cells are always dying. That is acceptable because cellular replication is also ongoing. Dead cells are replaced by new cells.

Cellular replication has TWO steps. Both must be named and distinguished on a test:

- **Proliferation:** cells increase in number (they proliferate).
- **Differentiation:** newly formed cells become different from each other — they specialize and develop unique capabilities.

Visual from the slide: proliferation is at the top (new cells being formed). Those new cells then pass through levels of differentiation until they become the mature specialized cells along the bottom. Mature cells have unique characteristics.

Mitosis vs Meiosis

	Mitosis	Meiosis
Which cells	Somatic cells = regular body cells	Germ cells → egg or sperm
What they become	All other cells in the body	Gametes only
Parent → daughters	1 parent → 2 daughter cells	Germ-cell replication
Chromosome # in new cells	46 chromosomes (same as parent). 23 PAIRS.	23 chromosomes — NOT 23 pairs. Just 23.
Why that number	Each daughter must match the parent exactly	Egg + sperm each bring 23 so the zygote has 46 (23 pairs). If each germ cell had 46, the zygote would have way too many.

DO NOT MIX UP: The instructor said “derm cells” once; she meant GERM cells. Germ cells reproduce by meiosis. Everything else reproduces by mitosis.

TEST TIP: Memorize 46 vs 23 exactly as worded: mitosis = 46; meiosis = 23 (not 23 pairs).

2. What Controls Proliferation

Three influences. Rank them the way the lecture ranked them.

1. Genetic programming — PRIMARY

It is genetically pre-programmed into each type of cell how often it should divide. Different cell types divide at different rates.

- **Liver cells:** normally divide only once every 1–2 years.
- **GI-tract epithelial cells:** may divide two or more times a day.

This is the primary influence on proliferation.

2. Cell density–dependent inhibition

Sounds fancy. All it means: cells proliferate until they contact a lot of other cells, feel crowded, and then proliferation slows.

Wound example: cells proliferate to fill a wound in the arm. When wound edges meet and cells feel crowded, that is the signal to slow proliferation because the wound is healed and extra proliferation is no longer needed.

> **TEST TIP:** Cancer cells LOSE cell density–dependent inhibition — they do not care if they get crowded. That is a high-yield contrast.

3. Situational factors

- Amount of growth factor present
- Amount of hormones available
- Amount of nutrients available
- Events such as blood loss or tissue injury — these can signal that proliferation needs to occur

Again: the PRIMARY determinant is what is programmed into that cell's genes.

3. The Cell Cycle

The cell cycle is the process the cell goes through as it proliferates. Four MAIN phases plus G0, which most people do NOT count as part of the cycle.

Phase	Stands for / name	What happens — exact lecture content
G1	Gap 1 — getting ready	Cell makes more proteins, more RNA, and more organelles (the cytoplasmic structures from last class). Needed because one parent must fully supply TWO daughter cells.
S	Synthesis	Synthesizing DNA. Parent must have enough DNA to supply two cells. Errors here → some genetic disorders.
G2	Gap 2	Pretty much a repeat of G1; similar preparation.
M	Mitosis	The cell actually splits into two. If chromosomes are not divided equally → chromosomal disorders.
G0	Resting phase — OUTSIDE the cycle	Most people do NOT consider G0 part of the cell cycle. Cell is out of the cycle. "Resting" means resting from replication only — the cell is still alive (makes ATP, etc.). It is not getting ready to divide and is not dividing.

> **TEST TIP:** If a question asks how many phases are in the cell cycle, the lecture's answer is FOUR (G1, S, G2, M). G0 is resting / outside the cycle.

Meiosis also has a cell-cycle-like process. Exact events in the phases differ from mitosis. Instructor will not test the meiosis-phase details at this point — only the general concept of a cell cycle plus the 46 vs 23 difference.

Why you care about the cell cycle (clinical)

Cancer development

- Cancer: HIGHER number of cells actively in the cell cycle replicating
- FEWER cells in the resting phase (G0) than there should be

- That produces uncontrolled cell growth / tumors getting bigger

Chemotherapy — cell-cycle specific drugs

Many chemo drugs are cell-cycle specific: they work during ONE phase and interfere with what is supposed to happen in that phase, so the cancer cell cannot replicate.

- **S-phase drugs:** prevent DNA from being replicated, often by interfering with enzymes needed for that process.
- **M-phase drugs:** keep the cancer cell from actually dividing into two.

This explains TWO prescribing facts:

- **Combination chemo:** often give more than one drug at a time — e.g., one S-phase + one M-phase — to wipe out more cancer cells than one drug alone.
- **Long duration:** must give the drug long enough for hopefully ALL cancer cells to cycle through that phase before replication can be halted.

Genetic / chromosomal disorders (preview)

- Something goes wrong in S phase → DNA replicated wrong → certain genetic disorders.
- Something goes wrong in M phase → chromosomes not divided equally between daughters → many chromosomal disorders.

4. Differentiation

Second step of cellular replication: new cells develop into different, more specialized kinds of cells.

The body has cells at varying levels of differentiation. Three levels were taught.

Level	Examples	Specialize?	Reproduce?	What they can make
Highly differentiated / specialized	Nerve cells; skeletal muscle cells; brain cells	YES — unique jobs (e.g., conduct action potentials)	MUCH less able to reproduce. Brain cells DO reproduce, but at a much, much slower rate (old belief: not at all).	Their own specialized type
Parent / progenitor cells (mid-level)	Skin cells; liver cells	Less fancy than nerve/muscle; cannot conduct action potentials	Much MORE capable of reproducing. Skin cells reproduce a lot / very often.	When they reproduce they produce only cells JUST LIKE THEM — only that one type.
Fairly undifferentiated (stem cells)	Bone-marrow stem cell	Few specialized capabilities; cannot do fancy things a nerve cell can do	Can enter the cell cycle and produce LARGE numbers of new cells when needed. Very good at replicating.	Usually more than one kind of cell. Bone-marrow stem cell → RBCs, platelets, AND all the different WBCs — not just another stem cell.

A very differentiated cell is NOT more valuable than an undifferentiated stem cell. You need BOTH: specialized function AND cells that can reproduce.

What controls how differentiated a cell becomes

PRIMARY: genetic programming — which genes are switched ON vs switched OFF.

Apparent paradox the instructor walked through (this is testable logic):

- Every cell comes from the original zygote.

- When a parent divides, each daughter is supposed to get the exact same genes / chromosomes / DNA.
- So every cell in the body pretty much has the SAME set of genes and chromosomes.
- How can genetic programming then decide which cells specialize?

Answer: in every cell some genes are “switched on” and some are “switched off.” Which genes are expressed determines how often the cell proliferates AND how differentiated it becomes.

Situational factors also affect differentiation (same family as proliferation): growth factors, nutrients, how much oxygen there is, whether cells get crowded. Still secondary. Primary = genetically programmed / which genes are expressed.

5. Cancer — Definition and Nature

Best-known problem when cellular replication goes wrong. BOTH steps are messed up: abnormal proliferation AND abnormal differentiation.

Cancer is not a single disease. Many kinds (skin, lung, bone, etc.) each with unique features. This lecture covers what almost ALL cancers have in common. Specific types come later in the curriculum.

Textbook definition used in class:

- Tissue growth that exceeds that of normal
- Not appropriate
- Serves no useful purpose
- In fact it is hurting the host
- And yet it is continuing

Current belief: cancer cells result from genetic mutations.

- **Mutation =** a change. Genetic mutation = a change in the person’s genes.

Logic chain she wants you to be able to recite: proliferation and differentiation are determined by genetic programming → change the genes (mutation) → change the programming → abnormal proliferation + abnormal differentiation = cancer.

6. Cancer Risk Factors

Most cancers are NOT due to just one factor but a COMBINATION. The more risk factors a person has, the more likely they are to get cancer. Different cancer types may relate more to different factors.

A. Chemical / environmental carcinogens

- **Carcinogen:** fancy word for something that can cause cancer.

Examples listed: radiation (including radiation from the sun), tobacco smoke, asbestos, toxic chemicals.

Mechanism from last recording: these generate free radicals. Free radicals steal electrons from cell parts and cause cell damage. If they steal electrons from genes/DNA → genetic mutation → depending on the mutation, predisposition to cancer.

- **Latency / time lag:** often 5 to 30 years between exposure and diagnosis. Instructor’s family example: uncle smoked ~2 decades, quit when she was 10, diagnosed with lung cancer when she turned 40 → ~30-year lag. Latency can make it hard to track the original risk factor, but if you know what to look for you can usually identify it.

B. Cancer-causing viruses (not all viruses)

A cold or the flu does NOT increase cancer risk. Only specific viruses linked to specific cancers:

- **HPV (human papilloma virus):** cervical cancer in women.
- **Hepatitis B and hepatitis C:** liver cancer.
- **Epstein–Barr virus:** certain kinds of lymphoma.

Why viruses can cause cancer (viral life cycle as drawn):

- Viruses are NOT like bacteria. Bacteria can replicate on their own outside body cells. Viruses cannot.
- Viruses must enter a host cell and use the host cell's machinery to replicate.
- Picture 1: virus enters host cell.
- Picture 2: viral DNA is incorporated into the host cell's DNA.
- Picture 3: host cell now has combination DNA (own + viral).
- DNA has the instructions for how to make protein → host now makes more copies of the virus. That is how the virus replicates.

Point for cancer: sticking viral DNA into host DNA IS a genetic mutation. Changed host genes → abnormal proliferation and differentiation → cancer.

C. Immune-system defects — immune surveillance hypothesis

Hypothesis: cancer cells are probably developing in everybody's body all the time. The immune system is supposed to recognize them as abnormal and get rid of them before they establish, grow, and cause problems. If the immune system is not working correctly, it will not recognize or eliminate cancer cells.

Support for the hypothesis:

- People with immunodeficiency disorders such as AIDS — higher incidence of cancer
- People on immunosuppressive drugs — higher incidence of cancer
- Older people — higher incidence of cancer, because as we age everything gets less efficient including the immune system

D. Heredity

A cancer-causing mutation in a GERM cell (egg or sperm) that then becomes a child: the child inherits the mutation and has increased cancer risk.

> **DO NOT MIX UP:** Mutation in a regular BODY (somatic) cell can cause a problem for that person but is NOT passed to kids. Only a germ-cell mutation is hereditary.

Example: a particular kind of breast cancer so highly hereditary that women who carry the mutation often have preventive mastectomies so they do not have to wait to see when they will get breast cancer.

7. Problems Caused by Abnormal Proliferation

In cancer:

- Higher percentage of cells actively in the cell cycle than usual
- Fewer cells resting than usual
- Loss of cell density–dependent inhibition — they do not care if crowded
- Tend not to die within their normal genetically pre-programmed time frame

Net result: uncontrolled cell growth.

Why extra cells are NOT “nice to have just in case”: cancer cells increase until they crowd out normal cells, steal the blood supply (and therefore oxygen and nutrients) from normal cells, and normal cells can no longer function.

8. Problems Caused by Abnormal Differentiation

Organ / tissue dysfunction

Cancer cells do not specialize properly, so they do not do what the normal cells of that tissue are supposed to do.

- Liver cancer cell ≠ normal liver-cell function → eventually liver failure
- Renal cancer cell ≠ normal renal-cell function → eventually renal failure

Expect dysfunction of whatever tissue or organ developed the cancer.

Lost cell-to-cell cohesiveness → metastasis

Normal cells like to stick together (cohesiveness). Cancer cells lose that. They do not mind breaking apart and wandering to other parts of the body. They spread through blood or lymph and start cancers there. That is metastasis.

Weird substances cancer cells make

1. VEGF — vascular endothelial cell growth factor

- **Vascular =** blood vessel
- **Endothelial cells =** cells that line blood vessels
- **Growth factor =** promotes growth

VEGF promotes growth of NEW blood vessels. Why that helps the tumor: a tumor can only get so big before it outgrows its blood supply (O2 + nutrients). If it can grow new vessels and keep increasing blood supply, it can keep growing.

2. Ectopic hormones

- **Ectopic =** in an abnormal location (same word as ectopic pregnancy: embryo not in the uterus, e.g., in the fallopian tube).

Ectopic hormones = hormones not produced by the endocrine glands as they should be, but by cancer cells.

Why this is a problem: endocrine glands are regulated by negative feedback so they do not overproduce. Cancer cells are NOT regulated by any feedback. They do not care how much hormone you already have. Result: hormone imbalance, usually TOO MUCH — patient looks like they have an endocrine disease.

Classic example: many LUNG cancers produce hormones just like ACTH, antidiuretic hormone (ADH), or parathyroid hormone (PTH). Unregulated overproduction → endocrine-disease picture.

3. Enzymes or toxins that damage cells

Tissue damaged by these often does not heal normally. That is why public-service “warning signs of cancer” lists include a sore that does not heal. Common with various skin cancers. Slide showed examples of sores that did not heal.

Abnormal antigens used as clinical markers

Many poorly differentiated cancer cells have abnormal antigens on their cell membranes. Used as clinical markers to MONITOR progression or EVALUATE treatment success — NOT to diagnose and NOT to screen.

Marker	Full name	Where found
PSA	Prostate-specific antigen	Cell membrane of some prostate-cancer cells
CEA	Carcinoembryonic antigen (lecture said “carcinombrionic”)	Certain GI cancers: colon, gastric, pancreatic; also certain breast cancers

Why NOT for diagnosis or screening:

- Not reliable for that purpose
- Some antigens appear in non-cancerous conditions
- Sometimes they do not appear until cancers are fairly advanced — and we want to find cancer before it is advanced

How markers ARE used — PSA walkthrough (memorize the logic)

- Man already diagnosed with prostate cancer by biopsy.
- Draw blood → measure PSA → start a chemo drug.
- Later redraw: PSA still high or higher → drug is NOT working → switch drugs.
- After new drug: PSA dropping → drug IS working → leave him on it.
- After remission: periodic blood draws (example: every 6 months for the first couple of years).
- Years later (example: 5 years) PSA starts rising again → maybe the cancer came back → bring him in and check it out.

TEST TIP: If a question asks “what is PSA/CEA used for?” the lecture answer is monitor progression / evaluate treatment / detect recurrence — never diagnose or screen.

9. Generalized Effects of Cancer

Cachexia

- **Cachexia =** weight loss; people wasting away. Advanced-cancer patients are usually very skinny.

Two reasons taught:

- **1. Nutrient theft:** cancer cells steal blood supply and nutrients from surrounding normal cells. Cancer grows; patient gets skinny.
- **2. Cytokine-induced anorexia:** immune system tries to kill cancer cells by releasing cytokines. Cytokine (working definition for now; more in a few weeks): a substance white blood cells make to help them do their job. Side effect of those cytokines: anorexia = loss of appetite. Patient is not hungry, so they lose weight. Family-brings-favorite-food scene: patient says “I can’t eat, I’m just not hungry.”

DO NOT MIX UP: Cachexia = the weight loss / wasting. Anorexia = the loss of appetite. They are related but not the same word.

Pain

Often one of the more challenging nursing issues. Adequate control may require enough medication that the patient becomes sedated. End-stage patients do not want to sleep away remaining time with family.

Reasons for cancer pain:

- Tumor presses on nerve endings
- Tumor presses on blood vessels in surrounding tissue → cuts off blood flow → tissues short of blood/O2 → pain
- Enzymes/toxins damage tissue → damaged tissue hurts
- Some experts think something about cancer directly causes release of chemical mediators of pain

Anemia

- **Anemia =** lower than normal RBCs or hemoglobin.

Reasons (cancer itself AND treatment):

- Anorexia → poor nutrition → iron deficiency or B-vitamin deficiencies
- Toxins damage tissue → damaged tissue often bleeds
- Growing tumors erode into blood vessels → bigger blood loss
- Metastasis to bone marrow → marrow cannot make RBCs
- Chemo or radiation can suppress bone marrow → anemia as a treatment side effect

If cancer is not controlled, people die. That is why the next distinction matters.

10. Benign vs Malignant Growths

If you had a tumor, you would rather have a benign one. Benign growths are not cancerous. Malignant growths are.

Cancer = BOTH proliferation AND differentiation abnormal. Benign = ONLY proliferation abnormal; differentiation is totally normal. So people with benign growths have roughly “half the problems” — problems from the mass getting bigger, not from cells that fail to work properly.

Feature	Benign	Malignant (cancer)
Which replication steps are abnormal	Proliferation only. Differentiation is normal.	BOTH proliferation and differentiation.
How it grows / spreads	Grows by EXPANSION only.	May grow by expansion AND spreads by METASTASIS. Infiltrates surrounding tissue.
Capsule	Very frequently surrounded by a fibrous capsule that contains it.	NO capsule. Sends out tentacles into normal tissue.
Surgery	Much easier to remove and feel confident you got all of it.	Hard to be confident every cell is out. Leave a few behind → they keep multiplying → cancer comes back.
Weird substances	Do NOT make VEGF, ectopic hormones, or damaging enzymes/toxins.	Can make VEGF, ectopic hormones, enzymes/toxins.
Systemic effects	Do not cause generalized effects like anemia or cachexia. Problems	Cachexia, anemia, pain, organ failure, metastasis, etc.

	are local, around the tumor.	
Death	Usually do NOT kill unless location or size interferes with a vital function.	Will kill if not controlled.

Exception she gave: certain BRAIN tumors that are benign can still kill because of location. When they get too big they squash brain tissue and interfere with regulation of breathing, heart rate, or blood pressure.

Slide images: malignant = lighter mass in the middle sending tentacles into yellowish normal tissue, no capsule. Benign = lighter mass with a capsule against reddish normal tissue, contained, easier to shell out.

11. Cancer Treatment — Logic, Not a Full Drug List

Surgery

Logic: remove the cancer → remove its harmful effects on nearby normal tissue (cannot steal nutrients/blood if gone). Limit: hard to be sure every cell is out; leftover cells multiply and the cancer returns. Therefore surgery is almost always COMBINED with radiation and/or chemo.

Radiation

Radiation generates free radicals. Goal: free radicals that either kill the cancer cell outright or damage it so it cannot replicate.

Downside: nearby normal cells can also be radiated.

Why it is still viable: cells most affected by radiation are the most rapidly proliferating cells — usually the cancer cells.

Chemotherapy

Many drugs work in a specific cell-cycle phase to block what should happen in that phase (enzymes, DNA, protein, or actual splitting into two).

Downside: usually PO or IV, so effects are not limited to cancer cells → almost always side effects.

Why still viable: most affected cells are the most rapidly proliferating — usually the cancer cells.

The irony she emphasized

Radiation and toxic chemicals/drugs are ON the list of things that can CAUSE cancer. Treating cancer with them can predispose to a different cancer later. Choice: without chemo/radiation the patient dies; with them you may buy extra time and accept later risk. Most people, after the risk is explained, still choose treatment. Instructor said she thinks she would too.

Newer drugs — biologic response modifiers / immunomodulators

Awareness slide only — you will hear these a lot later. They should not scare you.

- **Biologic response modifier:** a drug that modifies/changes some biologic response in the body — mostly immune responses. Makes the immune system more effective at getting rid of cancer.
- **Immunomodulator:** a drug that modulates/changes the immune system. Basically the same idea as a biologic response modifier.
- **Monoclonal antibody:** a man-made (cloned) antibody designed to target a specific substance causing a disease. Once you know the culprit substance, design an antibody to get rid of it.

- **Synthetic cytokines:** man-made versions of cytokines (substances WBCs make to help them do their jobs). Another way to change immune response and hopefully make it more effective.

12. Genetics Vocabulary — Different-Size Chunks of the Same Material

Up to this point DNA, genes, and chromosomes were used almost interchangeably. They are closely related but not the same. They are different SIZE chunks of genetic material.

DNA

- **DNA =** deoxyribonucleic acid.
- Complex molecule that stores genetic information.
- Made of chains of nucleotides.
- Nucleotides contain one of four possible nitrogenous bases. The bases carry the genetic information / form the genetic code.
- Bases shown as letters of the alphabet (A, C, G, T). Genetic code = which bases appear, in what order, and in what quantity.

From DNA to protein

- DNA is translated into RNA.
- RNA carries the genetic code out into the cytoplasm so ribosomes can use it to make proteins.
- The code is read in groups of THREE = a triplet code.
- **Each triplet = a codon.** Each codon represents ONE specific amino acid.
- Proteins are made of amino acids. Enough codons stuck together in order = instructions for how to make a protein (which amino acids, in what order, in what quantity).

Gene

When you put the codons together you get the instructions for one particular protein. THAT is a gene.

Used to think one gene = one protein. Now know some proteins are so big/complicated that more than one gene is needed. For this class: if you think “a gene is the instructions for how to make a particular protein,” you will not be far off.

Genes hooked end to end make one big DNA molecule.

Chromosome

Take that long DNA molecule and wind it really tight around a little protein core, then keep winding. Eventually you get a chromosome.

A chromosome = a DNA molecule wound up really, really tight. Many, many genes in each chromosome.

How many chromosomes

- Most body cells: 46 chromosomes.
- Arranged as 23 pairs.
- One chromosome of each pair from mom, one from dad.
- **22 of the 23 pairs = autosomes.** Alike whether you are male or female.

- **23rd pair = sex chromosomes.** Determine male or female. Males XY, females XX.

A karyotype with XY is a male genotype.

All genes in all those chromosomes together = genetic information for one unique person. Those 23 pairs contain instructions for all the different proteins the body has to make.

Nutshell + letter analogy (she used this twice — know it)

Letters analogy	Genetics equivalent
Letters of the alphabet	Nitrogenous bases on nucleotides (A, C, G, T)
Words (letters in order/quantity)	Codon = 3 bases = 1 amino acid
Sentences	Gene = instructions for one protein
Paragraph / story wound up tight	Chromosome = tightly wound DNA molecule with many genes
The whole book in one person	46 chromosomes in a cell = genetic information for one person

What genes DO

- **1. Regulate protein synthesis → control cell function.** Cells are made of proteins. Most cellular functions are carried out by proteins (enzymes, etc.). Genes have instructions for cell parts and for the enzymes/proteins needed to function. Tiny nutshell: genes control the function of the cell.
- **2. Determine inheritance patterns.** What traits pass from generation to generation — not only physical (hair color, eye color) but biochemical and physiologic traits (how fast/slow metabolic rate is; predisposition to diseases such as cancer).

13. Mendelian Inheritance

Basic patterns first proposed by Mendel. Mendel's laws published around 1865 (about the time of the Civil War). Still called Mendel's laws and still taught essentially as he proposed them.

Egg + sperm → zygote with 23 pairs of chromosomes (one of each pair from dad, one from mom). Because chromosomes are paired, there are TWO genes for every trait (one from mom, one from dad). Example: two genes for eye color; two for hair color.

- **Alleles:** the two members of a gene pair.
- **Dominant:** the more influential gene in the pair for determining the trait.
- **Recessive:** the other one.

If one dominant + one recessive → the DOMINANT gene is expressed. Classic: one brown-eye gene + one blue-eye gene → brown eyes, because brown is dominant for eye color.

Homozygous vs heterozygous

- **Homo = same.** Homozygous allele: both genes the same — both dominant or both recessive. Notation: uppercase A = dominant, lowercase a = recessive. AA or aa = homozygous.
- **Hetero = different.** Heterozygous: one dominant + one recessive (Aa).

Dominant gene expresses itself in EITHER a homozygous OR a heterozygous pairing.

Recessive gene expresses itself ONLY in a homozygous pairing — only if no dominant gene is present. Blue eyes require two blue-eyed recessive genes. One brown + one blue → brown eyes.

Carrier

A heterozygous person (e.g., brown-eyed with one brown + one blue gene) has brown eyes himself but CAN pass the blue-eye gene to children. He does not have blue eyes but carries blue-eyed genes. That person is a carrier.

Official definition used later: a carrier can pass on a disease but does not have symptoms themselves.

14. Two Kinds of Genetic Problems

Gene disorders vs chromosome disorders. Genes and chromosomes are both genetic material; chromosomes are the bigger chunk (many genes). A chromosome disorder therefore usually means a BIGGER NUMBER of kinds of problems — not necessarily more serious/severe, just more problems.

Gene disorders — origin

- Result of genetic mutations / a change in a person's genes.
- During meiosis, when DNA is being replicated, a mistake happens and DNA is not replicated accurately → mutation.
- Why? Sometimes radiation, toxic chemicals, or other environment. Often we never know. Textbook phrase: a "metabolic accident." Can occur spontaneously.

> **DO NOT MIX UP:** A mutation in a regular BODY cell can cause problems but is NOT called a gene disorder in this lecture. Gene disorders = mutation in a GERM cell, because that can be passed to the next generation. Those are the focus today.

Single-gene disorders

Mistake/mutation in ONE gene in a germ cell, then passed on. Multi-gene mistakes exist but inheritance is more complicated; this class only covers single-gene disorders (only one gene among all genes in all chromosomes is wrong).

Because a gene is instructions for a specific protein: that protein will be missing or defective/non-working.

- **Assessment findings:** depend on WHICH protein that gene codes for. NO blanket symptom list for all single-gene disorders. Some are very mild; some are fatal if the protein is crucial for normal cell function.
- **Interventions:** also no blanket statement. Geared toward treating the symptoms of that particular disorder. We do not currently know how to go in and fix the genetic mutation.

Single-gene disorders follow Mendelian patterns closely. Three primary patterns, determined by: (1) whether the defective gene is on a sex chromosome or an autosome, and (2) whether it carries a dominant or recessive trait.

15. Sex-Linked (X-Linked) Disorders

Defective gene is on a sex chromosome, not an autosome.

Less common than autosomal disorders: 22 pairs of autosomes vs only 1 pair of sex chromosomes, so a random gene mistake is more likely on an autosome.

When the defect is on a sex chromosome it is usually on the X, not the Y: there are far more X's than Y's. Everyone has an X (females two, males one). Only males have a Y. That is why sex-linked disorders are often called X-linked.

Most sex-linked disorders have a RECESSIVE inheritance pattern (instructor has never found a reference explaining why). Recessive gene is expressed only when no dominant/normal gene is present.

Why males have the disease and females are carriers

- Females have two X's. One defective recessive X is overridden by the normal dominant gene on the other X → female usually does NOT have the disease or symptoms. She can pass the defective X to kids → she is a carrier.
- Males have only one X. If that X is defective there is no second X with a normal gene to override it → male HAS the disease.

Could a normal gene on the Y override the bad X? No — Y is a lot smaller than X. X has more than a thousand additional genes that Y does not have. Quite possible there is no matching normal gene on the Y.

Pattern to memorize: males usually have the disease; females are usually carriers.

A female with two defective X's is theoretically possible but rarely happens. Instructor will not put that on the test.

Example: Hemophilia A — most common X-linked disorder

Defective gene = instructions for Factor VIII (factor 8), a clotting factor. Trouble with bleeding.

Slide: bruising; bleeding into a muscle; bleeding into a joint (joint swollen).

16. Autosomal Dominant Disorders

Defective gene on an autosome (sex does not matter) AND the defective gene carries a DOMINANT trait → it will be expressed.

- Only ONE defective gene is enough for symptoms / disease.
- There are NO carriers. Definition of carrier = can pass disease but has no symptoms. A person with one defective dominant gene can pass it AND has symptoms → they have the disease, they are not a carrier.
- To NOT have the disease you need two normal recessive genes (aa).

Notation: defective gene = uppercase A. Aa (or theoretically AA) = disease. aa = normal.

Example: Marfan syndrome

Defective gene has part of the instructions for making connective tissue. Connective tissue is all over the body → symptoms span multiple systems.

- **Musculoskeletal:** really long thin arms and legs; long thin fingers; chest and spinal deformities (scoliosis, "hunchback" appearance).
- **Eyes:** retinas become detached; lenses get dislocated because ligaments are weak (connective tissue in ligaments is weak).
- **Cardiovascular:** some heart valves prolapse; tendency to aneurysms.

How to diagram a parent who "has" an autosomal dominant disorder

Theoretically they could be AA (two defective uppercase genes). That is uncommon: it would require two people with the same autosomal dominant disorder to have a child and both pass a defective gene. That child would have a much more severe form. Most such couples choose not to have children.

> **TEST TIP:** On the TEST: if she says a parent has an autosomal dominant disorder, diagram them as ONE uppercase + ONE lowercase (Aa), not AA. She said if you ask her which to use she will tell you to diagram it the way the recording said: one uppercase, one lowercase.

17. Autosomal Recessive Disorders

Defective gene still on an autosome, but it carries a RECESSIVE trait. Recessive traits do not express if a dominant gene is present.

- To HAVE the disease: two recessive defective genes (aa). No dominant gene present.
- That means the child got a defective gene from BOTH parents. Both parents had a defective recessive gene and both passed it.
- Carriers exist: one normal dominant + one defective recessive (Aa). Dominant is expressed → no symptoms, but they can pass the defective gene.
- Not diseased and not a carrier: two dominant normal genes (AA).

Example: Sickle cell anemia

Defective gene = instructions for hemoglobin.

Striking detail she wants you to appreciate: not only is it one gene among tons of genes — that gene has only ONE amino acid coded incorrectly. All other amino acids are correct. That single amino-acid error makes defective hemoglobin.

Under certain conditions the defective hemoglobin makes RBCs become sickle-shaped. Sickled RBCs clump, form clots, occlude blood flow to parts of the body, and cause a lot of pain.

Sickled RBCs are not considered normal, so the body destroys them. Extra destruction → patient becomes anemic. Hence the name.

- **Hemolytic anemia:** hemo = blood / hemoglobin; lysis = breaking apart. Anemic because we are breaking down our own RBCs.
- **Two defective genes:** sickle cell anemia / sickle cell disease.
- **One defective gene (carrier):** sickle cell trait.

18. Punnett Squares — Test Rules

Used in genetic counseling to visualize a couple's risk of a child with a single-gene disorder or a carrier child. Draw boxes, put parent genes on the sides, fill in all possible combinations their children could get, calculate risk.

Some nurses do genetic counseling. She WILL put Punnett squares on the test. Students often struggle. Use her question list every time.

Resource she posted: module for Class Number Two, Cell Part Two — document titled "Help with Punnett Squares." No extra content; just organized in writing. Practice questions will also be done in class. Put specific questions on the pre-class exercise.

Questions to ask yourself on every problem

- Will you use X's and Y's, or uppercase and lowercase A's?
- Will you calculate risk out of TWO or out of FOUR?
- Is the defective gene dominant or recessive?
- How many defective genes do you need to have the disease? (May differ by sex in X-linked.)

- What combination of genes represents disease vs carrier vs normal? (This is where students most often fail — representing the PARENTS correctly.)

Master rule: 2 vs 4

	Sex-linked	Autosomal (dom or rec)
Letters used	X and Y (you CAN tell girls from boys)	Uppercase / lowercase A (you CANNOT tell girls from boys)
How she will word the question	"If this couple had a SON…" or "If this couple had a DAUGHTER…"	"If this couple had a CHILD…"
Denominator	Out of 2 (only the two possible kids of that sex). Risk = 0/2, 1/2, or 2/2.	Out of 4. Risk = 0/4, 1/4, 2/4, 3/4, or 4/4.

TEST TIP: Sex-linked → risk out of 2. Autosomal → risk out of 4. Getting the denominator wrong is the fastest way to miss the question.

How to write parent genotypes

Situation	Write it as	Notes
Male, normal genes (X-linked)	Normal X + Y	No defective X
Male WITH the X-linked disease	Defective X + Y	One bad X is enough
Female, normal genes (X-linked)	Two normal X's	
Female CARRIER (X-linked)	One defective X + one normal X	She does not have the disease
Anyone with an autosomal dominant disease (on her test)	Aa (one upper, one lower)	Do NOT use AA unless she specifically says two defective genes
Unaffected, autosomal dominant family	aa	Two normal recessive
Autosomal recessive DISEASE	aa (two lowercase)	Needs two defective genes
Autosomal recessive CARRIER	Aa	No symptoms
Autosomal recessive, normal / not a carrier	AA	Two dominant normal

Worked examples she did on the recording

X-linked: carrier mom + dad with normal genes

- Dad: normal X + Y. Mom: one defective X + one normal X.
- If they had a SON: risk he has the disease = 1/2 = 50%. That son has only one X; if it is the bad one he has the disease.
- If they had a DAUGHTER: risk she is a carrier = 1/2 = 50%. She has a normal X to override the bad one, so she does not have the disease.

X-linked: mom with normal genes + dad who HAS the disease

- Dad has the defective X. Sons get dad's Y, not his X → sons all have normal genes. Risk son has the disease = 0.
- Every daughter gets dad's (only) X, which is defective, plus mom's normal X → every daughter is a carrier. Risk daughter is a carrier = 2/2 = 100%.

> **TEST TIP:** X-linked inheritance pearl: affected dad cannot give the disease to his sons (they get his Y). He gives his defective X to ALL daughters, who become carriers if mom is normal.

Autosomal dominant: one parent has the disease (Aa) + other parent normal (aa)

Most common parental combination she will use. Risk a child has the disease = 2/4 = 50%. No carriers exist in this pattern.

Autosomal recessive

Many parental combinations are possible (normal / carrier / affected). Use the correct upper/lowercase combo for each parent, then calculate out of FOUR. Possible answers: 0%, 25%, 50%, 75%, or 100%.

19. Chromosome Disorders

A chromosome is a long DNA molecule with a whole ton of genes. A chromosome problem therefore usually means a greater number of kinds of problems than a single-gene disorder.

Important twist: in a chromosome disorder, all the genes that are present could be coded CORRECTLY and there would still be a problem — because of missing pieces or wrong NUMBER of chromosomes.

Broken-off / lost piece

During meiosis, when genes/chromosomes are being replicated, a piece may break off a chromosome and get lost. All the genes on that lost piece are missing → problems.

Nondisjunction (unequal split)

During meiosis the parent cell is supposed to divide chromosomes equally between two daughter cells. Parent makes two copies of every chromosome and gives one copy to each daughter.

This is meiosis, so each germ cell is supposed to have ONE copy of each chromosome. That germ cell then combines with another germ cell; each brings one copy; zygote ends up with TWO copies (the pairs you are supposed to have).

- **Disjunction:** the normal process — chromosomes divided equally.
- **Nondisjunction:** disjunction did NOT happen. Parent gives ONE daughter cell BOTH copies of that chromosome and the other daughter cell gets NONE.

Trouble appears when those daughter cells combine with other germ cells to make zygotes — wrong amount of chromosomes in the zygotes.

Monosomy

Daughter cell that got ZERO copies + a normal germ cell → zygote has only ONE copy of that chromosome (supposed to have two). Mono = one.

You do not see many people with monosomies walking around. Defects are usually pretty severe. Textbook: usually “incompatible with life” — fetus usually cannot survive and is usually miscarried.

Trisomy

Daughter cell that got BOTH copies + a normal germ cell → zygote has THREE copies. Tri = three. Supposed to have two.

Quite a lot of trisomies exist: trisomy 18, trisomy 13, trisomy 22. The number = which chromosome has the extra copy.

Most common: trisomy 21 = Down syndrome.

Down syndrome / trisomy 21

Karyotype: all chromosomes paired except chromosome 21, which has three copies (shown lower left on her slide).

The pictured child's karyotype on the slide was actually a MALE karyotype (XY) even though the photo was a girl — she used it so you could SEE that the Y is a lot smaller than the X (callback to the X-linked section).

Because of the extra chromosome, children have a constellation of physical and mental abnormalities, not one single-protein problem like sickle cell or hemophilia:

- Characteristic facial features
- Some degree of mental retardation
- Other possible defects: congenital heart problems, GI malformations, etc.

Assessment and treatment of chromosomal disorders

Same boat as single-gene disorders: no blanket assessment findings for all chromosomal disorders — problems depend on WHICH chromosome is involved. Interventions treat symptoms, which differ by chromosome. We do not know how to go in and fix the chromosome problem. Goal: treat symptoms and give the best quality of life possible.

You will study these more in peds, because these are problems babies are born with and that is when families learn to care for them and facilitate best development.

20. Rapid-Fire Test Bank (from her exact wording)

- Two steps of cellular replication: proliferation and differentiation.
- Mitosis = somatic / body cells, 46 chromosomes. Meiosis = germ cells, 23 chromosomes (not 23 pairs).
- Zygote chromosome math: 23 + 23 = 46 / 23 pairs. Germ cells start with 23 so the zygote is not overloaded.
- Primary control of both proliferation and differentiation = genetic programming (which genes are on/off).
- Liver divide 1–2 years; GI epithelium 2+ times/day.
- Cell density–dependent inhibition = contact/crowding slows division. Lost in cancer. Wound-edge example.
- Cell cycle phases: G1 (proteins/RNA/organelles), S (DNA synthesis), G2 (like G1), M (split). G0 = rest FROM replication, outside the cycle, cell still makes ATP.
- Cancer: more cells in cycle, fewer in G0.
- Chemo often cell-cycle specific (S vs M). Why combo drugs. Why long courses.
- S-phase errors → gene disorders. M-phase unequal split → chromosomal disorders.
- Highly specialized cells reproduce poorly (brain cells: slow, not zero). Progenitor cells (skin, liver) reproduce well but only make their own type. Stem cells (marrow) make many lineages (RBC, platelets, all WBCs).
- Same genome in (almost) every cell; expression (on/off) creates differences.
- Cancer = both steps abnormal; not one disease; growth exceeds normal, inappropriate, no useful purpose, harms host, continues. Result of mutations.
- Carcinogens (sun radiation, tobacco, asbestos, toxic chemicals) → free radicals → DNA electron theft → mutation. Latency 5–30 years.
- Viruses: HPV–cervical; HBV/HCV–liver; EBV–lymphoma. Viral DNA into host DNA = mutation. Viruses need host machinery; bacteria do not.

- Immune surveillance: cancers arise often; immune system should clear them. AIDS, immunosuppressants, aging ↑ cancer.
- Only germ-line mutations are hereditary. Example: highly hereditary breast cancer → preventive mastectomy.
- Abnormal proliferation: crowd out normals, steal blood/O2/nutrients.
- Abnormal differentiation: organ fails at its job; lose cohesiveness → metastasis via blood or lymph; VEGF (new vessels so tumor can grow past blood-supply limit); ectopic hormones (no negative feedback; lung ca → ACTH, ADH, PTH); enzymes/toxins → non-healing sores; abnormal antigens.
- PSA, CEA: monitor / judge treatment / watch recurrence. NOT diagnose, NOT screen. CEA: colon, gastric, pancreatic, some breast.
- Cachexia = wasting. Anorexia = no appetite (cytokine side effect). Two cachexia mechanisms: steal nutrients + cytokine anorexia.
- Pain: nerves, ischemia from vessel compression, tissue damage, possible pain-mediator release. Nursing tension: dose vs sedation vs family time.
- Anemia: nutrition, bleeding from toxins or vessel erosion, marrow mets, marrow suppression from chemo/radiation.
- Benign: proliferation only, expansion, often fibrous capsule, local problems, no VEGF/hormones/toxins/cachexia/anemia typically, rarely fatal except location (brain). Malignant: both steps, infiltrate + metastasize, no capsule, systemic effects.
- Surgery almost always combined with radiation and/or chemo because leftover cells regrow.
- Radiation = free radicals to kill or stop replication; hits fast dividers hardest.
- Treatment irony: radiation and toxic drugs are also carcinogens; still used because the alternative is death.
- Biologics / immunomodulators / monoclonal antibodies / synthetic cytokines — immune-modifying newer drugs.
- DNA → bases A,C,G,T → codon (triplet = one amino acid) → gene (one protein, mostly) → wound DNA = chromosome → 46 / 23 pairs. 22 autosome pairs + XY or XX.
- Genes control cell function via protein synthesis AND inheritance (physical + biochemical/physiologic).
- Mendel ~1865 / Civil War. Alleles. Dominant expressed in hetero or homo. Recessive only if homozygous. Carrier = hetero, no symptoms, can transmit.
- Gene disorder = germ-cell mutation (“metabolic accident”). Single-gene: one protein missing/bad; no universal s/sx or treatment; cannot yet fix the mutation.
- X-linked usually recessive, usually on X. Males affected, females carriers. Y too small (>1000 fewer genes). Hemophilia A = Factor VIII; bleeds into muscle/joint.
- Autosomal dominant: one bad gene = disease; no carriers. Marfan = connective tissue → long limbs/fingers, chest/spine deformity, lens dislocation / retinal detachment, valve prolapse, aneurysms. On test, affected parent = Aa.
- Autosomal recessive: aa = disease, Aa = carrier, AA = normal. Sickle cell: one wrong amino acid in hemoglobin gene → sickled RBCs → clots, pain, hemolytic anemia. Disease vs trait.
- Punnett: X-linked use X/Y and risk out of 2 by sex. Autosomal use A/a and risk out of 4. Affected X-linked dad + normal mom → 0% affected sons, 100% carrier daughters.
- Nondisjunction → monosomy (usually miscarriage / incompatible with life) or trisomy. Trisomy 21 = Down syndrome: face, intellectual disability, congenital heart, GI malformations. Extra chromosome = many problems even if each gene is “spelled” right.
- Chromosome disorders: treat symptoms / quality of life; revisited in peds.

21. Instructor Logistics (not content, but she said them)

- Submit questions as part of the pre-class exercise; she will try to answer them in class.
- More Punnett practice in class before the test.
- Textbook table has more examples of sex-linked, autosomal dominant, and autosomal recessive disorders if you want extra names.
- Chromosomal disorders will be revisited in peds.

End of recording: “That’s it for now. Remember to submit your questions… I will see you then. Bye.”